

NILESH BHAGAT

Senior Data Engineer | Healthcare Actuarial & Banking Analytics | Databricks | Azure | PySpark
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Summary

Senior Data Engineer with **12 years of IT experience** and **6+ years** in data engineering and business intelligence. Specialized in building scalable data pipelines for **healthcare and banking domains** using **Databricks, PySpark, Azure, and Delta Lake**. Hands-on experience in **Generative AI and RAG pipelines** — building enterprise knowledge retrieval systems using **LLMs, vector databases, and embedding models**. Experienced in actuarial reporting pipelines, credit risk models, and fraud detection systems. Skilled at working directly with business and technical stakeholders to deliver clean, reliable data within SLA windows.

Skills

Cloud & Platforms:: Azure Data Factory • AWS (S3, Redshift) • GCP (BigQuery) • Azure Databricks • ADLS Gen2 • Snowflake • Salesforce.CRM

Big Data & Streaming: Apache Spark • Kafka • Spark Streaming • Autoloader • Hadoop Ecosystem • OLAP • OLTP • Dynamics 365 CRM • EDA

Programming Languages: Python • PySpark • SQL • Pandas • T-SQL • Data Structures • SparkSQL • NoSQL • NumPy

DevOps & CI/CD: Airflow • ADF • Databricks Workflows • Logic Apps • Terraform • Git • Azure DevOps • Docker • GitHub Actions • MLOps

AI / GenAI: RAG • Vector Databases • LLM • LlamaIndex • CosmosDB • Embeddings & Vectorization • LLM Integration • OpenAI • Hugging Face

Data Modeling & Warehousing:

Star/Snowflake schema design • Dimensional modeling • Lakehouse architecture • SSAS • SQL Server • SSRS • Big Data • MongoDB, NoSQL • PostgreSQL

Other Tools: REST APIs • Power BI • Azure Purview • ERP & IoT • Metadata Management • Unity Catalogs • JIRA, Agile • Power Apps

PROFESSIONAL EXPERIENCE

LTIMindtree

Nagpur, India

Lead Data Engineer

07/2024 - Present

- Architect and build scalable, cloud-native data pipelines using Databricks, PySpark, ADF, Apache Spark, and Kafka to ingest medical, member, pharmacy, dental, and risk score data from Artemis for actuarial teams
- Designed and implemented end-to-end Databricks pipelines using Delta Live Tables, Autoloader, and Structured Streaming, reducing data ingestion latency by 45% across multiple production workloads
- Developed structured and unstructured data ingestion from multiple sources including APIs, databases, and streaming data sources
- Built automated pipelines to generate actuarial reports by aggregating claims, premium, and membership data from Artemis with full audit trail
- Designed and implemented RAG pipelines enabling enterprise knowledge retrieval using embedding generation, vector databases (ChromaDB), and LLM integration
- Built and enforced data quality frameworks using DQX, implementing validation rules, SLA monitoring, and automated alerting across 15+ pipelines — achieving 99.2% data reliability in production
- Collaborated directly with actuarial and underwriting teams to translate reporting requirements into scalable pipeline designs on Delta Lake

UST

Pune, India

Senior Data Engineer

09/2021 - 07/2024

- Developed ADF and Databricks pipelines for data ingestion and transformation for IoT devices using Spark and streaming frameworks.
- Built customer data pipelines on Databricks to consolidate banking transaction history, account profiles, and application data for lending and risk teams.
- Designed data pipelines to process and score personal loan applications — transforming real time raw customer financial data into risk-ready datasets for underwriting decisions.
- Stored and managed customer data in CRM systems — ensuring clean, consistent records across banking products for downstream reporting and analytics.
- Built reliable pipelines with logging, alerting, and governed tables to improve observability and data quality for CRM and banking systems.
- Mentored junior engineers, conducted code reviews, and supported onboarding of new team members.

Birlasoft

Pune, India

Data Engineer

01/2020 - 09/2021

- Migrated 3 on-prem SQL Server data warehouses to Azure Databricks lakehouse architecture, achieving 30% reduction in processing costs and 35% improvement in pipeline efficiency.
- Re-engineered legacy stored procedures into cloud-native PySpark jobs, cutting average runtime by 30% and eliminating single points of failure in batch processing.
- Implemented data models following star and snowflake schema best practices, ensuring analytics-ready dimensional structures aligned with business KPIs and BI standards.

PROFESSIONAL EXPERIENCE

Capita	Pune, India
ETL Analyst	07/2018 - 01/2020
- Validated and supported end-to-end ETL workflows, ensuring accurate extraction, transformation, and loading. - Developed automated validation frameworks using Python, SQL, and PySpark, significantly reducing manual testing effort. - Collaborated cross-functionally with BI teams to design analytics-ready data models for Power BI.	
Cognizant	Pune, India
Programmer Analyst	08/2016 - 07/2018
- Built automated ETL testing frameworks using PyTest and Selenium. - Integrated automation into CI/CD pipelines, cutting manual testing effort by 60%. - Support data quality checks: design validations, monitor data loads, investigate and fix data issues, and document resolutions.	
Infosys BPM	Pune, India
Software Engineer	07/2014 - 01/2016
- Improved data integrity by 25% through rigorous regression testing for ETL and schema changes. - Performed source-to-target verification, ensuring accuracy, completeness, and transformation correctness.	

Education

Nagpur University (RTMNU)	Nagpur, India
BE in Information Technology	03/2012

Certifications

Databricks Certified Data Engineer Associate — Databricks

Build RAG Applications — Coursera